

Static Gesture Classification — Run Report

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Data paths

Training data root	/home/jestin/ThesisRepo/ML/NewTrainingData/Static
External test root	/home/jestin/ThesisRepo/ML/NewTestData/Stephen/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats → feature dim 1408
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=680 · test=85

Augmentation (training only)

Enabled	yes
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (RBF)	0.9971	0.0088	1.0000
SVM (Linear)	1.0000	0.0000	1.0000
Random Forest	1.0000	0.0000	1.0000
Logistic Regression	0.9971	0.0088	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	0.9971	0.0088	1.0000
Stacking (SVM+RF+KNN+LogReg)	0.9971	0.0088	1.0000
KNN	0.9809	0.0175	0.9882
LDA	0.9338	0.0289	0.5176

Per-class hold-out F1

Class	SVM (RBF)	SVM (Linear)	Random Forest	Logistic Regression	Voting (soft)	Stacking	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.500	13
Double_Nothing	1.000	1.000	1.000	1.000	1.000	1.000	0.957	0.118	12
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.621	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.400	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.600	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	0.960	0.526	12
macro avg	1.000	1.000	1.000	1.000	1.000	1.000	0.988	0.490	85

External test (NewTestData)

Algorithm	External test acc
Voting Soft (SVM+RF+KNN+LogReg)	0.4929
Random Forest	0.4643
Stacking (SVM+RF+KNN+LogReg)	0.4643
SVM (Linear)	0.4500
Logistic Regression	0.4429
KNN	0.4214
SVM (RBF)	0.3071
LDA	0.1643

Per-class external accuracy

Class	Voting (soft)	Random Forest	Stacking	SVM (Linear)	Logistic Regressi on	KNN	SVM (RBF)	LDA	Support
Double_Closed_Fist	0.100	0.100	0.050	0.000	0.050	0.050	0.000	0.350	20
Double_Nothing	0.800	1.000	0.850	0.850	0.300	0.550	0.950	0.000	20
Double_Okay	0.050	0.000	0.050	0.000	0.000	0.150	0.000	0.200	20
Double_Open_Palm	0.650	0.700	0.600	0.400	0.700	0.650	0.350	0.100	20
Double_Phone	0.550	0.550	0.550	0.800	0.750	0.100	0.350	0.250	20
Double_Pistol	0.650	0.000	0.500	0.450	0.600	0.800	0.050	0.200	20
Double_Spiderman	0.650	0.900	0.650	0.650	0.700	0.650	0.450	0.050	20